

# **Education Under Fire**

Conflict Exposure and Educational Attainment in Northern Nigeria

Economics Department Undergraduate Honors Thesis

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## Abstract

This paper examines the effect of violent conflict on the educational attainment and school enrollment of children in Nigeria. Using data from the 2010 Nigeria Education Data Survey (NEDS) and geocoded violence data from the Uppsala Conflict Data Program (UCDP) spanning 1999–2010, I estimate OLS models with and without LGA fixed effects to identify whether exposure to violence at different stages of childhood affects the probability of school enrollment and the number of years of schooling completed. I find that exposure to violence during early childhood (ages 0–3) and preschool age (3–6) is associated with a reduced likelihood of current school enrollment, while the effects on total years of schooling are mixed and sensitive to model specification. Results confirm well-documented gender and urban-rural disparities in educational attainment. These findings motivate continued investment in school access and protection of education during conflict.

**Keywords:** conflict, education, Nigeria, Boko Haram, school enrollment, UCDP, OLS, fixed effects

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## I. Introduction

Close to one-third of all nations have experienced some form of internal conflict in the past 40 years, with roughly twenty percent experiencing a conflict lasting more than ten years (Blattman and Miguel, 2010). Moreover, the poorest nations—those in Sub-Saharan Africa, Asia, Eastern Europe, and South America—tend to have the highest propensity for war and violent conflict. These conflicts devastate the lives, health, and living standards of the people who experience them, even though there has been no conclusive evidence that in the long run these episodes of violence affect the development trajectory of a country (Davis and Weinstein, 2002; Miguel and Roland, 2011).

The purpose of this paper is to measure the effects of violence on educational attainment among school-age children in Nigeria. I use survey data from the 2010 Nigeria Education Data Survey (NEDS) and combine this with violence data spanning 1999–2010 from the Uppsala Conflict Data Program (UCDP) to examine whether experiencing violence in a child’s local government area (LGA) affects the number of years of education a student will attain and whether it increases the likelihood that a student will be enrolled in school.

The history of Nigeria is one marked by violence and unrest since the country gained independence from Great Britain in 1960. This legacy has been attributed to the fact that Nigeria’s borders, like those of most former African colonies, were drawn arbitrarily without consideration of the tribal or ethnic relations of those who would soon call themselves fellow countrymen. Nigeria’s case is particularly complex due to its high degree of ethnic and tribal diversity: there are more than 300 different ethnic groups within its borders. Major struggles for control and power arise as tribes fight for control of natural resource-rich lands, political parties compete for control of those resources, and religious groups attempt to impose their ideologies on members of other religious communities.

There is a growing literature that attempts to measure the role violence plays in educational attainment and economic returns to schooling. In general, it has been found that violence

has a negative effect on educational attainment and that children exposed to violence earlier in life attain less schooling than their unexposed peers (León, 2012; Shemyakina, 2011). Experiencing violence also appears to have a larger negative effect on girls than on boys (Shemyakina, 2011). I review this literature in Section 3.

The remainder of this paper is organized as follows. Section 2 provides a history of education in Nigeria. Section 3 reviews the existing empirical literature on the determinants of school dropout and the effects of violence on educational attainment. Section 4 describes the datasets used in this paper. Section 5 details the econometric methodology. Section 6 presents and discusses the results. Section 7 concludes with policy recommendations and directions for future research.

## II. Education in Nigeria

The history of modern education in Nigeria began in the mid-nineteenth century, when Christian missionaries introduced the “Western” style of education to the region (nig, Yeara). These new methods of schooling, along with Western bureaucratic and social norms, were quickly adopted by those living in the southern region of the country—notably in the coastal towns of Lagos, Calabar, and Port Harcourt—producing some of the first African professionals, civil servants, lawyers, and doctors. In contrast, the predominantly Muslim North was far less welcoming of these new methods and ideas. Northern communities rejected the Christian missionaries and the idea of Western education out of concern that it would interfere with their faith, and they sought to create an alternative that emphasized scholarship in Islamic studies. The result was the creation of *Islamiyya* schools—established and operated by Muslim scholars with instruction grounded in a core Muslim legal and religious curriculum (Umar, 2003).

Around the time of independence and in the years following, the Nigerian state took an active role in expanding access to education for all citizens, recognizing the importance of an educated population to the country’s development goals. The first generation of Nigerian

universities were established between 1948 and 1965 to meet the growing demand for an educated citizenry. These institutions were fully funded by the federal government, which held that it was in the national interest to ensure that all students had equal opportunity to attend school regardless of socioeconomic status (Babs Fafunwa, 1992). In time, as demand for education continued to grow, more schools at all levels—primary, secondary, and tertiary—were built across the country.

The 1970s oil boom played a major role in expanding the education sector, given Nigeria's position as a major oil producer. Following the Nigerian- Biafran Civil War of 1967–1970, and in response to growing inequality in educational attainment and economic outcomes between the North and the South and across gender lines, the federal government devised the **Universal Primary Education (UPE) scheme** in 1976 (Csapo, 1983). The UPE followed a 6-3-3-4 system: six years of primary school, three years of junior secondary school, three years of senior secondary school, and four years of tertiary education. The scheme aimed to ensure universal primary school completion by making primary education free and compulsory.

With these provisions, primary school enrollment grew well beyond what policymakers had initially projected—a problem compounded by the fact that initial enrollment estimates were based on dubious population data from the 1963 census (Csapo, 1983). The resulting oversubscription led to a deterioration of school resources and a decline in the quality of the UPE system. The number of schools was not keeping pace with the growing population, there was a chronic shortage of qualified teachers, and those teachers who were employed were not regularly paid (Babs Fafunwa, 1992). These challenges were only exacerbated when oil revenues—and thus government funding for education—declined in the 1980s. Mismanagement and corruption meant that only a fraction of the budgeted funds were actually directed toward educational projects. By 1985, the federal government had effectively ceased funding primary education (Umar, 2003). The continued deterioration of the education sector into the late 1990s meant that graduates were not adequately equipped with the skills they were

meant to acquire in school, contributing to massive unemployment across the country.

As a remedy, the Federal Government introduced the **Universal Basic Education (UBE) scheme** in 1999, again with the aim of providing greater access to quality basic education, and in line with the United Nations Millennium Development Goals. The UBE scheme set forth the following objectives:

- Providing uninterrupted access to nine years of formal education
- Reducing school dropout
- Making primary education free and compulsory
- Ensuring the acquisition of literacy, numeracy, and life skills
- Instilling a value of education in the populace (Universal Basic Education Commission, 2013)

The Universal Basic Education Commission (UBEC) adopted as its philosophy the “integration of the individual into sound and effective citizenry, and to provide equal education and opportunities for all.” The new scheme follows a 9-3-4 system.<sup>1</sup> Following nine years of free and compulsory primary and junior secondary school, students sit the National Common Entrance Examination (NCEE) to determine eligibility for admission to senior secondary school.<sup>2</sup> Following senior secondary school, students sit the Secondary School Certificate Examination and may apply to tertiary institutions.

Despite these policy efforts, deep regional and gender disparities in educational attainment persist. Northern Nigeria continues to lag behind the South on virtually every educational

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<sup>1</sup>The 9-3-4 system consists of nine years of free and compulsory education—six years of primary school and three years of junior secondary school—followed by three years of senior secondary school and four years of tertiary education. The UBE promotes automatic promotion through primary grades while ensuring quality through continuous assessment.

<sup>2</sup>Each state has approximately two federal government colleges funded and managed by the federal government through the Ministry of Education. Admission is merit-based (NCEE results) with nominal tuition of approximately 16,000 naira. State-owned secondary schools, funded by individual states, are generally of lower quality. Private schools charge 160,000–320,000 naira annually but offer smaller classes, modern equipment, and more highly qualified teachers (nig, Yearb).

indicator, a pattern that the Boko Haram insurgency—which began in earnest in 2009—has dramatically intensified, as discussed in subsequent sections.

### III. Literature Review

Given the role education plays in poverty reduction, productivity growth, and development, questions about the effectiveness of educational investments are of fundamental importance to policymakers in both developed and developing countries. One issue that has received sustained attention in this literature is school dropout—or early exit from school—and its determinants.

Several factors beyond simple inability to pay have been shown to drive dropout. Lack of school quality, or the *perceived* lack thereof, plays an important role (Levy, 1971; Hanushek et al., 2008; Patron, 2008). Hanushek et al. (2008), studying primary school-age children in Egypt, find that students are more likely to drop out when attending low-quality schools, holding all other factors constant. Patron (2008) similarly finds that perceived quality is an important determinant of secondary school attendance in Uruguay. Levy (1971) points to additional factors including high grade repetition rates, elevated social tension, and high fertility rates as negative drivers of educational attainment. For older primary students, the economic return to schooling and its opportunity cost also matter: as O’Higgins et al. (2008) show in the case of Italy, dropout decisions can be entirely rational and may occur even in the absence of binding monetary constraints.

The primary concern of this paper, however, is the relationship between violence and educational attainment—a topic that has been widely studied and which generally points in the same direction. Exposure to violence has been found to have a negative effect on educational attainment across a range of settings and conflict types (Akresh and de Walque, 2008; Swee, 2009; Rodríguez and Sánchez, 2012; Shemyakina, 2011; León, 2012).

Akresh and de Walque (2008) study the Rwandan genocide and compare the educational outcomes of children exposed to the violence with those who were not. They find that exposed



children achieved 0.5 fewer years of schooling and were 15 percent less likely to complete third grade. Shemyakina (2011) examines the civil war in Tajikistan and finds that school-age girls exposed to violence were the most negatively affected cohort relative to unexposed girls and boys of both exposure statuses. León (2012) finds that children exposed to violence at a very young age—before starting school—attained 0.31 fewer years of education than unexposed children, and that early exposure has a persistent effect, whereas older children exposed to violence after starting school eventually catch up with their non-exposed peers.

Not all studies find uniformly negative effects at all margins. Swee (2009), studying the Bosnian war, finds that greater war intensity reduced the probability of completing secondary school but had no effect on primary school completion rates. He attributes reduced secondary enrollment to demand-side effects, specifically the increased likelihood of military conscription. Rodríguez and Sánchez (2012) examine Colombian children aged 6–17 and find that children older than 11 are most affected by conflict through induced early labor market entry, mediated by increased mortality risk, negative income shocks, and reduced school quality.

Taken together, this literature establishes that the relationship between violence and schooling is negative in expectation but heterogeneous in magnitude across age of exposure, gender, conflict type, and outcome measured. This paper contributes to the literature by applying these frameworks to the Nigerian context, where violence has been pervasive but where large-scale, nationally representative education data offer an opportunity to estimate effects across a diverse population spanning multiple ethnic, religious, and geographic contexts.

## IV. Data

### *A. Nigeria Education Data Survey (NEDS)*

The primary data for this paper come from the 2010 Nigeria Education Data Survey (NEDS). The survey was conducted by the National Population Commission (NPC), the Federal Ministry of Education (FMOE), and the Universal Basic Education Commission (UBEC), with technical assistance from RTI International and joint funding from the United States Agency for International Development (USAID) and the UK Department for International Development (DFID). The objectives of the 2010 NEDS include (National Population Commission, 2010):

- Providing data on the schooling status of Nigerian children of basic education age, including factors influencing whether children ever enroll in school and why students drop out
- Measuring parent attitudes toward schooling quality and providing an understanding of the attitudes that shape willingness to send children to school
- Measuring the frequency of student absenteeism and reasons for missing school
- Providing data that allow for trend analysis and state-level comparisons

The survey targeted children aged 4–16 with a focus on the household-level factors affecting education consumption decisions. Data collection was conducted across all 36 states and the Federal Capital Territory (Abuja) from April to July 2010. In total, 26,934 households were interviewed, yielding information on 71,567 children.

### *B. Uppsala Conflict Data Program (UCDP)*

The violence data were collected by Uppsala University as part of the Uppsala Conflict Database Program. The program’s purpose is to provide comprehensive event-level data

on all organized violence since 1989. An event is defined as “any incidence where there is a use of armed force by an organized actor against another organized actor or civilians, which results in at least one direct death at a specific location and for a specific temporal duration” (Sundberg et al., 2010).<sup>3</sup> Uppsala University maintains accuracy in the dataset through a four-step data collection process involving systematic searches of major newswires and databases, review of secondary materials, consultation with regional experts, and final review by program directors.<sup>4</sup>

The dataset records date(s), event type, violence type, location details (town, province, and latitude/longitude coordinates), actors involved, and deaths for each side as well as civilian and unknown deaths. Events are classified as single-day, summary, or continuous events. Violence types are categorized as state-based, non-state, or one-sided violence. For this study, I use all events recorded in Nigeria from 1999 to 2010.

### *C. Linking the Datasets*

In order to link the household survey data with the violence data, I matched records using state and LGA identifiers present in both datasets. Some observations in the violence dataset did not contain LGA-level information and were therefore excluded from the matched sample. Summary statistics for the key variables used in the analysis are presented in Table 1.

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<sup>3</sup>See Appendix A.3 for complete UCDP definitions of key terms including organized actor, direct death, and event types.

<sup>4</sup>See Appendix A.2 for a description of the four-step data collection process.

**Table 1:** Summary Statistics

| Variable                                       | Mean | Std. Dev. | Min | Max |
|------------------------------------------------|------|-----------|-----|-----|
| <i>Panel A: Outcomes</i>                       |      |           |     |     |
| Currently enrolled ( <i>attend</i> )           |      |           | 0   | 1   |
| Years of schooling ( <i>highest_enrolled</i> ) |      |           | 0   |     |
| <i>Panel B: Violence Exposure</i>              |      |           |     |     |
| Violence: pre-birth (−17 to −9 months)         |      |           | 0   | 1   |
| Violence: in utero (0–9 months pre-birth)      |      |           | 0   | 1   |
| Violence: early childhood (ages 0–3)           |      |           | 0   | 1   |
| Violence: preschool (ages 3–6)                 |      |           | 0   | 1   |
| Violence: primary school (ages 6–12)           |      |           | 0   | 1   |
| Violence: junior secondary (ages 12–15)        |      |           | 0   | 1   |
| Violence: senior secondary (ages 15–18)        |      |           | 0   | 1   |
| <i>Panel C: Individual Controls</i>            |      |           |     |     |
| Male                                           |      |           | 0   | 1   |
| Age                                            |      |           |     |     |
| Urban                                          |      |           | 0   | 1   |
| Wealth quintile                                |      |           | 1   | 5   |
| Distance to primary school (km)                |      |           |     |     |
| Mother attended school                         |      |           | 0   | 1   |
| Father attended school                         |      |           | 0   | 1   |
| Observations                                   |      | [N]       |     |     |

*Notes:* Summary statistics for the analysis sample. Violence indicators are binary variables equal to 1 if the child’s LGA experienced at least one UCDP-recorded violent event during the specified age window. Violence exposure is assigned based on the child’s LGA of residence and the start date of the first UCDP event in that LGA. See Section 5 for details on variable construction.

## V. Methodology

In this paper I estimate the effect of community-level violence exposure on two distinct educational outcomes: current school enrollment and cumulative years of schooling. I define educational attainment as the number of years of school a student has completed.

### A. *School Enrollment (attend)*

To capture the effects of violence on school enrollment, the dependent variable is *attend*—a binary indicator equal to one if the student is currently enrolled in school and zero otherwise. The baseline estimating equation is:

$$attend_i = \alpha + \beta \mathbf{violence}_i + \gamma \mathbf{student}_i + \delta \mathbf{X}_i + \varepsilon_i \quad (1)$$

where  $\mathbf{violence}_i$  is a vector of binary indicators for whether the child’s LGA experienced at least one UCDP-recorded violent event during each of the following age windows: pre-birth (17 to 9 months before birth), in utero (0 to 9 months before birth), early childhood (ages 0–3), preschool (ages 3–6), primary school age (ages 6–12), junior secondary age (ages 12–15), and senior secondary age (ages 15–18).

$\mathbf{student}_i$  is a vector of individual-level student characteristics including age, age-squared, gender, religion, an indicator for late school start, Quranic school attendance, Tsangaya school attendance, Islamiyya school attendance, disability indicators (seeing, hearing, speaking, mobility, mental, other), and distance to primary, junior secondary, and senior secondary schools.  $\mathbf{X}_i$  contains additional household and community-level controls: mother’s and father’s living status, whether each parent resides in the household, parental educational attainment, community wealth quintile, region indicators, and the number of eligible school-age children in the household.

I estimate Equation (1) by OLS with heteroskedasticity-robust standard errors, and I also estimate an augmented version that includes LGA fixed effects to absorb unobserved

time-invariant LGA-level characteristics:

$$attend_i = \alpha + \beta \mathbf{violence}_i + \gamma \mathbf{student}_i + \delta \mathbf{X}_i + \mu_\ell + \varepsilon_i \quad (2)$$

where  $\mu_\ell$  denotes LGA fixed effects.

**Violence variable construction.** The violence binary indicators in  $\mathbf{violence}_i$  were created using the UCDP event start dates and each child’s estimated age. The household survey provided month and year of birth for each child; I assigned a birth date of the 15th of the reported birth month. I then calculated the difference between each child’s birth date and the start date of the first recorded UCDP event in their LGA, if the LGA experienced any violence, to determine the child’s age at first community violence exposure. A child was coded as not having experienced violence in a given window if born more than 17 months after the first recorded event in their LGA, since we do not expect violence that occurred well before a child’s birth to affect that child. The 17-month pre-birth window additionally allows estimation of the effects of a mother’s prenatal living conditions on child outcomes.<sup>5</sup>

**Late school start.** The variable *late*—a binary indicator for whether a student began school after reaching age six—was constructed from survey questions asking about reasons for late school entry, rather than from the age-at-entry variable in the original dataset, which contained implausible values (including ages younger than the student’s current age).

**Quranic, Tsangaya, and Islamiyya variables.** These three variables were present in the original dataset but contained missing data for a large share of observations. To address this, I assumed that only Muslim children are likely to attend one of these school types. Accordingly, for children who identified as Christian, Traditionalist, or other non-Muslim religion, all three variables were set to zero. This resolved most missing values; the remaining missing observations were retained as missing.

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<sup>5</sup>See Appendix A.1 for a complete description of all constructed variables and their definitions.

**Disability variables.** The original disability variable was a string in which each disability category was represented by a letter. I decomposed this string into six separate binary indicators: *seeing*, *hearing*, *speaking*, *mobility*, *mental*, and *dis\_other*.<sup>6</sup>

**Parental education.** The variables on mother’s and father’s educational attainment contained missing values for many observations, typically because the level of schooling reached was recorded but the attendance indicator was missing. For observations where the school level variable was non-zero, I imputed school attendance as one. The variables *mom\_enrol* and *dad\_enrol*—total years of schooling—were constructed by summing years implied by the level and class reached (e.g., if a parent reached secondary school level 2, class 3, years of enrollment = 6 + 3 = 9). This approach does not account for grade repetition, which the survey does not record for parents.

### ***B. Educational Attainment (highest\_enrolled)***

To capture the effects of violence on cumulative educational attainment, I use the variable *highest\_enrolled*, which measures the total years of schooling completed. The estimating equation parallels Equation (1):

$$highest\_enrolled_i = \alpha + \beta \text{violence}_i + \gamma \text{student}_i + \delta \mathbf{X}_i + \varepsilon_i \quad (3)$$

with an augmented version including LGA fixed effects.

**Construction of *highest\_enrolled*.** The outcome variable was constructed to reflect the maximum years of schooling completed by each student across three sources of information in the dataset: (1) current class and school level for currently enrolled students, (2) class and level from the previous school year (2008–2009), and (3) highest class and level reached for students who had dropped out. For currently enrolled students, the current class was assigned

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<sup>6</sup>The string was split into six single-character variables; a binary indicator for each disability was then set to 1 if the corresponding character appeared in the original string.

as the highest class reached. For each school level and class combination, I converted to years of schooling by summing the years required to reach that level (e.g., a student at senior secondary school level 1, class 3 has completed  $6 + 3 + 3 = 12$  years). I then took the maximum across all three sources to handle data entry errors, as a number of currently enrolled students recorded a current grade lower than the previous year’s grade. This approach assumes the highest recorded grade is the true grade reached, which may introduce upward bias for a small number of observations.

## VI. Results

### A. School Enrollment

Table 2 presents OLS estimates of Equation (1) (Column 1) and Equation (2) with LGA fixed effects (Column 2) for the *attend* outcome.

Looking at the baseline OLS results (Column 1), the violence indicators statistically significant at conventional levels are exposure during early childhood (ages 0–3), preschool age (3–6), and primary school age (6–12). The coefficient on *viol\_ech* is  $-0.022$  ( $p < 0.01$ ), on *viol\_ps* is  $-0.049$  ( $p < 0.001$ ), and on *viol\_prim* is  $+0.019$  ( $p < 0.01$ ). These estimates indicate that children exposed to violence at younger ages are less likely to be currently enrolled in school, while children who experienced violence during primary school age exhibit a counterintuitive positive association with current enrollment. The remaining violence variables carry positive coefficients except for *viol\_SS*, which is negative, though none are statistically significant. Pre-birth violence exposure shows no statistically significant effect on current enrollment.

When LGA fixed effects are included (Column 2), the model fit improves substantially (R-squared increases from 0.43 to 0.52), and several previously insignificant violence variables become significant. Exposure during the pre-birth window (17–9 months before birth) is now positive and significant at the 5% level ( $\hat{\beta} = 0.031$ ), and in-utero exposure is positive and



significant at the 1% level ( $\hat{\beta} = 0.039$ ). The early childhood effect is no longer statistically significant and the coefficient falls to near zero. The primary school age effect remains positive and significant, with a larger coefficient (0.055) than in the baseline model.

Across both specifications, male students are significantly more likely to be enrolled in school (coefficient  $\approx 0.04$ – $0.05$ ,  $p < 0.001$ ). The urban indicator is small and insignificant in the baseline but becomes larger and significant with fixed effects ( $\hat{\delta} = 0.035$ ), as expected once we absorb time-invariant LGA-level confounders. All parental education variables are positive, and mother's and father's alive status are not statistically significant with LGA fixed effects. Distance to primary school has a statistically significant negative effect on enrollment; distance to junior secondary and senior secondary schools is not significant.

**Table 2:** OLS Estimates: Effect of Violence Exposure on School Enrollment

|                                     | Dependent variable: <i>attend</i> |                     |
|-------------------------------------|-----------------------------------|---------------------|
|                                     | (1) OLS                           | (2) OLS + LGA FE    |
| <i>Violence exposure indicators</i> |                                   |                     |
| Pre-birth (−17 to −9 months)        | 0.004<br>(0.014)                  | 0.031**<br>(0.015)  |
| In utero (0–9 months pre-birth)     | 0.016<br>(0.013)                  | 0.039***<br>(0.014) |
| Early childhood (0–3 years)         | −0.022***<br>(0.007)              | 0.010<br>(0.010)    |
| Preschool age (3–6 years)           | −0.049***<br>(0.007)              | −0.017<br>(0.010)   |
| Primary school (6–12 years)         | 0.019***<br>(0.007)               | 0.055***<br>(0.011) |
| Junior secondary (12–15 years)      | 0.018<br>(0.019)                  | 0.032<br>(0.023)    |
| Senior secondary (15–18 years)      | −0.019<br>(0.074)                 | −0.115<br>(0.105)   |
| Individual controls                 | Yes                               | Yes                 |
| LGA fixed effects                   | No                                | Yes                 |
| Observations                        | 56,878                            | 56,878              |
| R-squared                           | 0.428                             | 0.517               |

*Notes:* OLS estimates with robust standard errors in parentheses. Individual controls include age, age-squared, gender, urban/rural, region, wealth quintile, parental education and living status, Quranic/Tsangaya/Islamiyya school attendance, late school start, disability indicators, and distances to school.  $*p < 0.10$ ,  $**p < 0.05$ ,  $***p < 0.01$ .

### ***B. Educational Attainment***

Table 3 presents results for the *highest\_enrolled* outcome—total years of schooling completed.

In the baseline OLS (Column 1), nearly all violence regressors carry negative coefficients, with the exceptions of *viol\_ps* (preschool,  $\hat{\beta} = 0.154$ ,  $p < 0.01$ ) and *viol\_JS* (junior secondary,  $\hat{\beta} = 0.052$ , not significant). All violence variables except *viol\_prim*, *viol\_JS*, and *viol\_SS* are statistically significant.

The LGA fixed effects results (Column 2) tell a markedly different story. After absorbing LGA-level confounders, all violence indicators except *viol\_SS* exhibit positive associations with educational attainment, and most are statistically significant at the 5% level or better (*viol\_uter\_p* and *viol\_ech* are not significant). This reversal in sign across specifications suggests substantial negative selection of violence into low-performing LGAs: once we compare children within the same LGA, violence exposure is associated with more—not fewer—years of schooling. This result is difficult to interpret causally and points to important identification challenges discussed in Section 7.

Male students achieve more years of schooling than female students across both specifications (coefficient  $\approx 0.014$ , significant). Urban children achieve significantly more schooling, and the urban premium triples in magnitude with the introduction of LGA fixed effects. Similarly, the coefficient on *region* increases from approximately 0.06 to 0.21 with fixed effects, underscoring the importance of geographic controls.

**Table 3:** OLS Estimates: Effect of Violence Exposure on Years of Schooling

|                                     | Dependent variable: <i>highest_enrolled</i> |                  |
|-------------------------------------|---------------------------------------------|------------------|
|                                     | (1) OLS                                     | (2) OLS + LGA FE |
| <i>Violence exposure indicators</i> |                                             |                  |
| Pre-birth (−17 to −9 months)        |                                             |                  |
| In utero (0–9 months pre-birth)     |                                             |                  |
| Early childhood (0–3 years)         |                                             |                  |
| Preschool age (3–6 years)           | 0.154***                                    |                  |
| Primary school (6–12 years)         |                                             |                  |
| Junior secondary (12–15 years)      | 0.052                                       |                  |
| Senior secondary (15–18 years)      |                                             |                  |
| Individual controls                 | Yes                                         | Yes              |
| LGA fixed effects                   | No                                          | Yes              |
| Observations                        | [N]                                         | [N]              |
| R-squared                           |                                             |                  |

*Notes:* OLS estimates with robust standard errors in parentheses. See Table 2 notes for description of individual controls. Full regression output is in Appendix B. \* $p < 0.10$ , \*\* $p < 0.05$ , \*\*\* $p < 0.01$ .

## VII. Conclusion

The purpose of this paper was to determine whether a link exists between community-level violence and educational attainment and school enrollment in Nigeria. Using the 2010 NEDS household survey and the UCDP violence dataset for 1999–2010, this analysis does not find a clear, consistent causal link between violence in a child’s LGA and household education

consumption decisions. However, two important patterns emerge.

First, the analysis confirms several well-established findings from the broader NEDS survey. Men achieve 1.5 more years of schooling than women in urban areas and 1.8 more years in rural areas, and primary school attendance is 64% for boys versus 58% for girls (National Population Commission, 2010). These gender gaps are confirmed in our regressions, where the coefficient on *male* is positive and statistically significant across all specifications. Urban children are also consistently more likely to attend primary school (74% vs. 55%) and secondary school (60% vs. 36%), a pattern borne out by the positive and generally significant coefficient on *urban*.

Second, the reversal in sign of the violence coefficients when moving from the baseline OLS to the LGA fixed effects specification—particularly for the *highest\_enrolled* outcome—highlights a fundamental identification challenge. Violence is not randomly assigned across LGAs. In Nigeria, LGAs that experienced violence during 1999–2010 differ systematically from those that did not in ways that correlate with educational outcomes. The OLS results absorb some of this selection through covariates but likely not all of it. The fixed effects estimates eliminate time-invariant LGA-level confounders but cannot address potential simultaneity or time-varying confounders. Cleaner causal identification—perhaps through an instrumental variables strategy or a difference-in-differences design exploiting geographic proximity to conflict sites—would be an important avenue for future work.

### ***A. Policy Implications***

Although a clear causal link between violence and educational attainment was not established, the results do point to several actionable policy directions.

The persistent urban-rural gap in school attendance, driven in large part by distance to school, calls for expanded geographic access to schooling in rural communities. Survey data indicate that 85% of urban children live within 15 minutes of their school, compared with 62% of rural children. In the northern region, average walk times to school range from 23 to

37 minutes, versus 14 to 19 minutes in the South (National Population Commission, 2010). Distance was reported by parents as one of the primary reasons children started school late or did not attend at all. Targeted school construction programs in underserved areas could meaningfully reduce this barrier.

Monetary costs also remain a binding constraint. Even though the UBE scheme makes primary and junior secondary school free, families must pay for uniforms, books, and supplies out of pocket. Better management of school funds and potentially subsidized supply provision could reduce this burden for the poorest households.

### ***B. Limitations and Future Research***

The most important limitation of this analysis is the absence of a credible identification strategy for isolating the causal effect of violence. Future work should pursue one of the following approaches:

1. **Difference-in-differences** exploiting the onset of the Boko Haram insurgency in 2009 as an exogenous shock, comparing pre- and post-2009 educational outcomes across high- and low- exposure LGAs using multiple DHS rounds (2008, 2013, 2018).
2. **Instrumental variables** using geographic or conflict- specific variation in violence exposure that is plausibly exogenous to educational outcomes.
3. **Richer data** with complete and reliable violence records at the LGA level and individual-level panel data to track educational trajectories over time.

The data limitations encountered here—particularly missing values, imputed variables, and erroneous data entry—also point to the importance of investing in higher-quality administrative and survey data collection infrastructure in Nigeria. Without reliable data, researchers cannot reach strong conclusions, and policymakers cannot design effective interventions.

## References

- ([Year]a). Nigeria education fact sheet. [Add URL or source].
- ([Year]b). Nigeria education profile. [Add URL or source].
- Akresh, R. and de Walque, D. (2008). Armed conflict and schooling: Evidence from the 1994 rwandan genocide. (4606).
- Babs Fafunwa, A. (1992). *History of Education in Nigeria*. NPS Educational Publishers, Ibadan.
- Blattman, C. and Miguel, E. (2010). Civil war. *Journal of Economic Literature*, 48(1):3–57.
- Csapo, M. (1983). Universal primary education in nigeria: Its problems and implications. *African Studies Review*, 26(1):91–106.
- Davis, D. R. and Weinstein, D. E. (2002). Bones, bombs, and break points: The geography of economic activity. *American Economic Review*, 92(5):1269–1289.
- Hanushek, E. A., Lavy, V., and Hitomi, K. (2008). Do students care about school quality? determinants of dropout behavior in developing countries. *Journal of Human Capital*, 2(1):69–105.
- León, G. (2012). Civil conflict and human capital accumulation: The long-term effects of political violence in Perú. *Journal of Human Resources*, 47(4), 991–1022, 2012.
- Levy, V. (1971). The distributional impact of government expenditures and revenue: With applications to kenya. [Update with correct Levy 1971 citation from thesis].
- Miguel, E. and Roland, G. (2011). The long-run impact of bombing vietnam. *Journal of Development Economics*, 96(1), 1–15, 2011.

- National Population Commission (2010). 2010 nigeria education data survey technical report. Technical report, National Population Commission, Abuja, Nigeria.
- O'Higgins, N. et al. (2008). [verify full o'higgins et al. 2008 citation from thesis].
- Patron, R. (2008). [verify full patron 2008 citation from thesis references].
- Rodríguez, C. and Sánchez, F. (2012). Armed conflict exposure, human capital investments and child labor: Evidence from colombia. *Defence and Peace Economics*, 23(2):161–184.
- Shemyakina, O. (2011). The effect of armed conflict on accumulation of schooling: Results from tajikistan. *Journal of Development Economics*, 95(2):186–200.
- Sundberg, R., Lindgren, M., and Pads kocimaite, A. (2010). Ucdp ged codebook version 1.0–2011. *Department of Peace and Conflict Research, Uppsala University*.
- Swee, E. L. (2009). On war and schooling attainment: The case of bosnia and herzegovina. *Households in Conflict Network Working Paper*.
- Umar, M. S. (2003). Education and islamic trends in northern nigeria: 1970s–1990s. *Africa Today*, 48(2):127–150.
- Universal Basic Education Commission (2013). About ubec. <http://www.ubec.gov.ng>.



## I. Data Appendix

### A. Variable Definitions

Table 4 defines all variables used in the analysis.

**Table 4:** Variable Definitions

| Variable                | Definition                                                                                        |
|-------------------------|---------------------------------------------------------------------------------------------------|
| <b>viol_uter_p</b>      | Binary. Experienced violence in LGA 17 to 9 months before birth.                                  |
| <b>viol_utero</b>       | Binary. Experienced violence in LGA 0 to 9 months before birth (in utero).                        |
| <b>viol_ech</b>         | Binary. Experienced violence in LGA ages 0–3 (early childhood).                                   |
| <b>viol_ps</b>          | Binary. Experienced violence in LGA ages 3–6 (preschool).                                         |
| <b>viol_prim</b>        | Binary. Experienced violence in LGA ages 6–12 (primary school).                                   |
| <b>viol_JS</b>          | Binary. Experienced violence in LGA ages 12–15 (junior secondary).                                |
| <b>viol_SS</b>          | Binary. Experienced violence in LGA ages 15–18 (senior secondary).                                |
| <b>attend</b>           | Binary. Equal to 1 if student is currently enrolled in school.                                    |
| <b>highest_enrolled</b> | Continuous. Total years of schooling completed (constructed from school level and class reached). |
| <b>mom_enrol</b>        | Continuous. Mother’s years of schooling.                                                          |
| <b>dad_enrol</b>        | Continuous. Father’s years of schooling.                                                          |
| <b>dist_JSS</b>         | Continuous. Distance in kilometers to the junior secondary school the child attends.              |

*Continued on next page*

Table 4 (continued)

| Variable         | Definition                                                                                                                                        |
|------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>dist_SSS</b>  | Continuous. Distance in kilometers to the senior secondary school the child attends.                                                              |
| <b>dist_prim</b> | Continuous. Distance in kilometers to the primary school the child attends.                                                                       |
| <b>late</b>      | Binary. Equal to 1 if student began school after reaching age 6. Constructed from survey questions on reasons for late school entry. <sup>a</sup> |
| <b>age2</b>      | Continuous. Age-squared.                                                                                                                          |

<sup>a</sup> The dataset contained an age-at-entry variable that appeared erroneous in some cases (e.g., reported ages younger than the student’s current age, or reported starting age consistent with late entry despite no late-entry reason being given). The *late* variable was constructed from a series of survey questions asking the student to indicate why they started school late. See Section 5 for details.

### ***B. UCDP Data Collection Process***

The Uppsala Conflict Data Program (UCDP) maintains accuracy and comprehensiveness through a four-step data collection process (Sundberg et al., 2010):

1. An online search is conducted using Factiva, an online database carrying more than 10,000 newswires, newspapers, and other sources. For each country, at least one major newswire (Reuters, AFP, Xinhua, EFE) and BBC Monitoring are searched using conflict-specific terms. Results are coded according to UCDP criteria.
2. Other available materials are searched and coded, including recently published books, case studies, journals, NGO publications, and other online databases.

3. Regional experts are consulted to clarify remaining ambiguities.
4. Finally, UCDP project managers and program directors review the data and determine what is included in the final dataset.

### ***C. UCDP Violence Definitions***

**Armed force** Use of arms to promote a party's general position in conflict, resulting in deaths.

**Arms** Any material means, including manufactured weapons or sticks, stones, fire, water, etc.

**Organized actor** One of the following: (a) the government of an independent state (the party controlling the capital); (b) a formally organized group with an announced name using armed force against a government, another formally organized group, or civilians; (c) an informally organized group without an announced name that uses armed force against similarly organized groups in a clear pattern of connected violent incidents.

**Direct death** A death directly related to combat between warring parties or against civilians.

**Civilians** Unarmed populations that are not part of the organized actors.

#### **Event types:**

- **Single-day:** A single incident of armed violence, or separate incidents resulting in at least one fatality occurring within one 24-hour calendar day at a specified location. Multiple incidents at the same location on the same day involving the same actors may be aggregated.
- **Summary:** Deaths from multiple separate events over several days for which disaggregated information is unavailable.

- **Continuous:** Armed activities at a specified location covering multiple interrelated days of continuous fighting.

**Violence types:**

- **State-based:** Conflicts between two organized groups where at least one is the state government.
- **Non-state:** Conflicts between two organized armed groups where neither is the state government.
- **One-sided:** Conflicts between an organized actor and civilians.

## II. Full Regression Output

Tables 5–?? present the complete regression output including all control variables. The main text reports only the violence exposure coefficients for brevity.



**Table 5:** Full Output: OLS Regression for *attend* (No LGA FE)

| Variable                   | Coefficient | Std. Error | <i>p</i> -value |
|----------------------------|-------------|------------|-----------------|
| <i>Violence indicators</i> |             |            |                 |
| viol_uter_p                | 0.0036      | 0.0136     | 0.789           |
| viol_utero                 | 0.0156      | 0.0129     | 0.226           |
| viol_ps                    | −0.0494     | 0.0071     | 0.000           |
| viol_prim                  | 0.0186      | 0.0070     | 0.008           |
| viol_JS                    | 0.0175      | 0.0193     | 0.364           |
| viol_SS                    | −0.0193     | 0.0743     | 0.795           |
| viol_ech                   | −0.0217     | 0.0068     | 0.001           |
| <i>Individual controls</i> |             |            |                 |
| age                        | 0.1309      | 0.0024     | 0.000           |
| age <sup>2</sup>           | −0.0060     | 0.0001     | 0.000           |
| male                       | 0.0463      | 0.0028     | 0.000           |
| urban                      | 0.0030      | 0.0036     | 0.400           |
| region                     | 0.0131      | 0.0009     | 0.000           |
| qhech3                     | −0.0009     | 0.0008     | 0.275           |
| mom_alive                  | 0.1303      | 0.0824     | 0.114           |
| mom_HH                     | 0.0002      | 0.0054     | 0.971           |
| mom_attend                 | 0.0987      | 0.0058     | 0.000           |
| mom_enrol                  | 0.0030      | 0.0005     | 0.000           |
| dad_alive                  | 0.0585      | 0.1233     | 0.635           |